

# Seyed Navid Hashemi Jazi

AI & Computer Vision Engineer

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## Education

- 2022–2026 **B.S. in Computer Engineering (Graduated)**, *University of Tehran*, Tehran, Iran, GPA: 18.31/20 (Rank: 12/81)  
Granted **Direct Admission to Master's Program** (Exceptional Talents quota) without entrance exam.
- 2018–2022 **High School Diploma, Physics & Mathematics**, *Imam Khomeini Exemplary High School*, Shahin-Shahr, Iran  
Ranked 144th in National University Entrance Exam;

## Professional Experience

- 2024–2025 **Computer Vision Intern**, *Rahtab Co. (Livestock AI)*, Tehran, Iran
- Developed **cattle biometric identification system** using deep metric learning (triplet loss, contrastive loss) on nose patterns
  - Compared state-of-the-art architectures: EfficientNet-B5, ResNet50, and Custom CNN; optimized for deployment
  - Achieved **95.9% ID prediction accuracy** and **92% known/unknown classification accuracy** on 603 validation images
  - Engineered Persian web UI for production deployment and real-time inference
- 2024 **Computer Vision Intern**, *Rahtab Co. (Agricultural AI)*, Tehran, Iran
- Built automated **broiler chicken counting pipeline** using YOLOv8 instance segmentation and object detection
  - Trained on custom dataset (162 test images, 845 instances) handling high-density and partial occlusion scenarios
  - Achieved **98% mAP@0.5** with real-time inference (~15-20ms/image on GPU) for production poultry management
  - Implemented confidence filtering, batch processing, and comprehensive evaluation metrics

## Key Projects

- Project **ReputationRadar: LLM-Powered Sentiment Dashboard**, *Python, Streamlit, OpenAI API, Docker*
- Architected end-to-end **real-time brand monitoring dashboard** aggregating sentiment from Reddit, Twitter/X, and Trustpilot
  - Implemented AI-powered sentiment classification using OpenAI with intelligent VADER fallback for reliability
  - Engineered exportable reports (PDF, Excel, CSV) with executive summaries using ReportLab and Plotly visualizations
  - Deployed with Docker, pytest coverage, and production-grade error handling & logging

- Project **Web Scraper & Data Analyzer with AI Chat**, *Python, PyQt5, OpenAI, OpenCV*
- Built full-stack **modern PyQt5 GUI application** for multi-threaded web scraping with advanced loop prevention
  - Integrated **conversational AI assistant** for website analysis, SEO insights, and content recommendations
  - Implemented dual preview system (text + HTML rendering), real-time progress tracking, and multi-format export (JSON, Excel, PDF)
  - Features: URL normalization, duplicate detection, interactive data visualization, configurable crawl depth (1-100 levels)
- Project **Deep Generative Models Research**, *PyTorch, Python, Google Colab*
- Implemented and studied **4 major generative model families**:
    - **Diffusion Models**: DDPM, DDIM, Stable Diffusion with DreamBooth/LoRA fine-tuning
    - **Flow Matching**: Novel continuous generative model for financial time-series with Wasserstein Distance evaluation
    - **Score-Based Models**: NCSN with Annealed Langevin Dynamics and class-conditional generation
    - **Energy-Based Models**: Contrastive Divergence training with denoising capabilities
  - Advanced theoretical understanding: manifold hypothesis, score matching, density estimation, representation learning
- Project **CPY Language Compiler (3-Phase)**, *Java, ANTLR, Maven*
- Developed **complete compiler infrastructure** with parser/AST generation, name analysis, and type checking
  - Implemented **Dead Code Elimination** optimization and **memory leak detection** security analysis
  - Engineered symbol table management with scope analysis and comprehensive error reporting
- Project **5-Stage Pipelined RISC-V Processor**, *Verilog, Digital Design*
- Designed and implemented **32-bit, 5-stage pipelined RV32I processor** from scratch in Verilog
  - Engineered complete **Hazard Detection Unit** with data forwarding paths to resolve pipeline conflicts

## Technical Skills

|                           |                                                                                                            |
|---------------------------|------------------------------------------------------------------------------------------------------------|
| <b>Languages</b>          | Python, C++, Java, JavaScript, C, Verilog, SQL                                                             |
| <b>AI &amp; ML</b>        | PyTorch, TensorFlow, Scikit-learn, Transformers, OpenAI APIs                                               |
| <b>Vision &amp; Data</b>  | YOLOv8, OpenCV, NumPy, Pandas, Matplotlib, Plotly, Albumentations                                          |
| <b>Specializations</b>    | Generative Models (Diffusion, VAE, Flow Matching, EBM), Computer Vision, Metric Learning, LLM Applications |
| <b>DevOps &amp; Tools</b> | Docker, Git, Streamlit, PyQt5, Google Colab, Linux/Unix, ANTLR, Maven                                      |

## Languages

|         |                                  |
|---------|----------------------------------|
| Persian | Native                           |
| English | Intermediate (Reading & Writing) |

## Certifications & Honors

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|------|-----------------------------------------------------------------------------------------|
| 2024 | Direct Admission to Master's Program (Exceptional Talents quota) — University of Tehran |
| 2022 | National Entrance Exam Rank: 144th nationwide                                           |